

Towards Balanced Information Propagation in Social Media

Abstract

As people increasingly rely on social media platforms such as Twitter to consume information, there are significant concerns about the low diversity of news consumption. Users may be exposed to posts which reinforce their pre-existing views, which could lead to a more fragmented society. For example, one worry is reinforcing societally damaging misinformation prevalent in one subgroup, such as COVID-19 denial. Aiming to combat this, earlier work divided news stories into *high consensus* and *low consensus* posts, based on the degree to which reactions can be expected to be similar from users with different political views. In this work, we propose and quantify the benefits of a strategy to *spread* high consensus news across readers with diverse political leanings. We first compile a dataset and make the following three key observations: (1) low consensus news is more likely to remain within subgroups of users with similar political leanings, whereas high consensus news spreads broadly to many users across subgroups; (2) high consensus news posted by neutral publishers spreads widely and more equally across subgroups; and (3) users that get the information from other users instead of the publishers get an even more biased exposure to news. We propose a strategy that spreads high consensus news through neutral publishers, and quantify the significant decrease in the disparity of users' news exposure. To scale and modernise the consensus detection pipeline, we further introduce an LLM-based zero-shot classifier that replaces costly crowdsourced annotation, and show that it faithfully reproduces the original findings on the 2017 dataset. Applying the same classifier to a fresh corpus of political news collected in 2026 across the same ten publishers confirms that the core pattern — neutral publishers produce substantially more high-consensus content than partisan outlets — remains stable nearly a decade later. Our extensive experiments on Twitter show that seeding high consensus information with neutral publishers is an effective way to achieve high spread to many users with little disparity regarding political leaning, and our updated results demonstrate that this conclusion generalises beyond the original news cycle.

Consensus; News consumption in social media; Filter bubble; Large language models; Political polarization.

Introduction

People increasingly rely on social media platforms, such as Facebook and Twitter, to receive news and information

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(Lichterhan 2010; Teevan, Ramage, and Morris 2011; Newman et al. 2023). Many news stories are divisive, often posted by polarized publishers, eliciting different reactions from users with different political leanings or pre-existing views, e.g., conservatives or liberals. While various news sources publish high and low consensus news that cover a given story, users often limit themselves to the low consensus divisive stories which can reinforce their prior views (Himmelboim, McCreery, and Smith 2013; Babaei et al. 2018; Chiang and Knight 2011; Robertson et al. 2018). This selective exposure and consumption of divisive information may lead to a more politically fragmented, less cohesive society (Cha et al. 2010) and the formation of filter bubbles or echo chambers (Bakshy, Messing, and Adamic 2015b; Bozdag 2013; Flaxman, Goel, and Rao 2016; Pariser 2011; Guess et al. 2023).

For example, only 12% of Conservative Fox News viewers believe that climate change is man-made. This compares with 60% of Americans overall, recognizing that 28% of Conservatives don't watch Fox News.¹ Americans who relied on Fox News, or similar right-wing sources, often do not believe the coronavirus threat is serious.² Social media platforms such as Facebook and Twitter can amplify segregation in information consumption with recommendation algorithms that incentivise people towards more extreme positions (Bakshy, Messing, and Adamic 2015a; Huszár et al. 2022; Haroon et al. 2022).

Due to concerns about information segregation and societal polarization (Sunstein 2002; Schkade, Sunstein, and Hastie 2007; Settle 2018), prior works have proposed exposing users to diverse news stories by nudging users to read other views (Munson, Lee, and Resnick 2013; Park et al. 2009; Balayn et al. 2023). While this could be useful for encouraging debate in society, such approaches have been shown to *increase* the chance that users reject stories from other perspectives — perhaps because they believe other publishers and their stories are biased — thereby defeating the purpose (Bail 2014; Lord, Ross, and Lepper 1979; Nyhan and Reifler 2010; Taber and Lodge 2006; Wood and Porter 2019; Bail et al. 2018; Levy 2021).

However, social media platforms receive so many posts that

¹<https://www.insider.com/fox-news-republican-viewers-reject-climate-change-science-2019-3>

²https://www.washingtonpost.com/lifestyle/media/the-data-is-in-fox-news-may-have-kept-millions-from-taking-the-coronavirus-threat-seriously/2020/06/26/60d88aa2-b7c3-11ea-a8da-693df3d7674a_story.html

some form of moderation is inevitable.³ We present one proposal to be considered: highlighting high consensus news that elicits similar responses from both sides could act as a soothing balm, helping people to connect with others, even if they have ideological differences. Babaei et al. (2018) proposed such a complementary approach to increase diversity in users’ information consumption by identifying high consensus, yet interesting information. Their system recommends high consensus “purple” posts to both “red” (conservatives) and “blue” (liberals) users, hoping to increase users’ exposure to cross-cutting news posts, leading to lower societal polarization and lower *segregation* in information consumption (Chakraborty et al. 2017; Eady et al. 2019).

Nevertheless, it still remains unclear how such information is spread across users in a network and how individuals choose to react to it. Vraga and Bode (2017) claim that if users receive even one piece of news from reliable news sources about health, it has a significant effect on them. Recent work further shows that exposure to authoritative, cross-cutting information can meaningfully reduce misperceptions even in highly polarized environments (Pennycook and Rand 2021; Altay, Berriche, and Acerbi 2023). Therefore, exposing people to high consensus news related to important events such as COVID-19 may improve health and harmony across society.

In this paper, we investigate users’ willingness to share and spread such posts, along with the reach of high and low consensus news stories across a diverse audience. We also examine the newsworthiness (Galtung and Ruge 1965; Weber 2014) of both high and low consensus news. Fundamentally, we ask if we can help to propagate high consensus stories broadly across society, and whether this finding holds beyond the original news cycle. We hope this can help to break undesirable echo chambers, leading to a less polarized, healthier society.

Note that due to the huge number of posts, social media platforms must inevitably engage in some form of story curation or promotion. We propose one approach which we believe can benefit society. The problem of *identifying* high consensus news automatically has already been addressed by Babaei et al. (2018), who relied on costly crowdsourced annotation via Amazon Mechanical Turk. In this work, we additionally show that a modern large language model (LLM) can replace this annotation step entirely, classifying news posts as high or low consensus in a zero-shot manner at negligible cost, while faithfully reproducing the original distributional findings (Gilardi, Alizadeh, and Kubli 2023; Törnberg 2023; Ziems et al. 2024). Recent work has demonstrated that LLMs match or exceed crowd-worker performance on a wide range of political and social science annotation tasks, from stance detection to framing analysis (Heseltine and Lasser 2024; Mellon et al. 2024; Alizadeh et al. 2023; Wang and Waseem 2021). This removes the primary bottleneck that prevented scaling the detection pipeline beyond a fixed dataset, and enables us to validate the core empirical patterns on a fresh corpus of political news collected in 2026 — confirming that neutral publishers produce substantially more high-consensus content than partisan outlets nearly a decade after the original study. The stability of this pattern across time is consistent with recent findings on the persistence of partisan editorial tendencies in online news (An et al. 2023; ?). We assume that platforms are good at selecting stories which are likely to prove popu-

lar, hence we focus on identifying a good set of stories from which to select. We highlight the following contributions:

- I. We compile a novel dataset, which reveals how Twitter users with similar or different political leanings are connected to each other. To do so, we consider a dataset of 400 news tweets posted by 10 publishers containing 100 high and 100 low consensus posts (Babaei et al. 2018). For every high or low consensus news post, we collected a subset of its 100 random retweeters and for each retweeter we collected a random set of their 100 followers. We compute the political leaning of the 1,616,000 users who either retweeted a high or low consensus news story or were exposed to it. We collected 40 high and low consensus news related to COVID-19 and global warming. Moreover, to simulate the spread of news in Twitter, we crawl a network of more than 100 million Twitter users (Kwak et al. 2010). This allows us to compute the political leanings of 69,687 users connected by 2,907,026 links.
- II. Using our dataset, we study how individuals with different political leanings get exposed to and retweet high and low consensus news posted by users from various political perspectives. We observe that low consensus news tends to proliferate primarily only amongst users with a particular political leaning (Cinelli et al. 2021; Del Vicario et al. 2016). Gruzd and Mai (2020) show how one single tweet on COVID-19 denialism circulated among many conservatives. In contrast, high consensus news has a higher chance of spreading through the entire network. Importantly, high consensus news posted by a set of neutral publishers spreads more equally across liberal and conservative users than if posted by the same number of a mix of non-neutral publishers.
- III. Based on the above observations, we propose a strategy that seeds neutral publishers to expose roughly equal fractions of people with different political leaning to high consensus news with the minimum cost, drawing on fair influence maximization (Ali et al. 2022; Tsang et al. 2019a; Fish et al. 2019). We show that our proposed strategy is more effective than seeding the most influential nodes without taking the political leanings into account.
- IV. We introduce an LLM-based zero-shot consensus classifier that replaces the original crowdsourced annotation pipeline (Gilardi, Alizadeh, and Kubli 2023; Alizadeh et al. 2023), and demonstrate that it reproduces the distributional findings of the original study on the 2017 dataset. Applying the same classifier to a new corpus of ~1,000 political news items collected in 2026 across the same publishers confirms that the core pattern — neutral publishers produce substantially more high-consensus content (61%) than conservative (45%) or liberal (42%) outlets — remains stable nearly a decade later (Ziems et al. 2024; Mellon et al. 2024), providing strong out-of-time empirical corroboration of the paper’s central claim.

Our work provides new insights and a complementary tool which may help to reduce echo chambers, encourage healthier interaction between population subgroups, and lead to a more cohesive society.

Related Work

We review related work on diverse and polarized news dissemination, LLM-based text annotation, and information propagation in social networks.

³<https://onezero.medium.com/the-moderation-war-is-coming-to-spotify-substack-and-clubhouse-9fe00672091b>

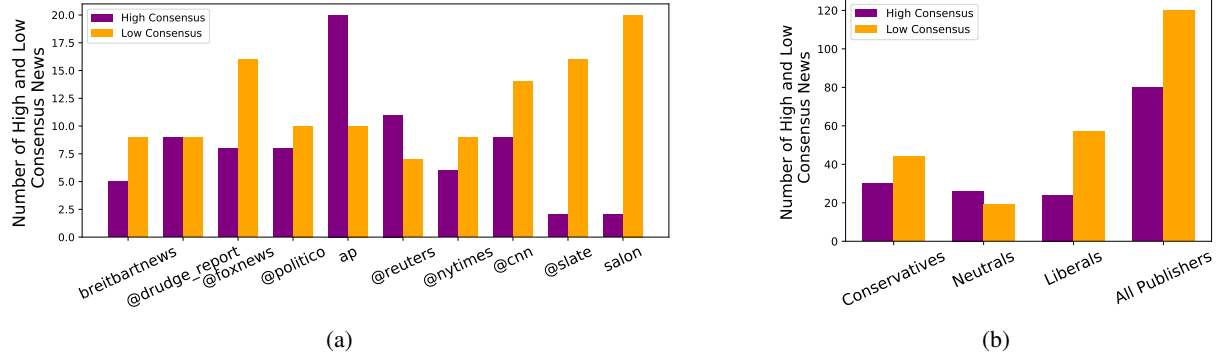


Figure 1: Number of high and low consensus news posted on Twitter during 9th-15th May, 2017. (A) shows the number of high and low consensus news for 10 selected publishers, and (B) shows the aggregated result for conservative, liberals, and neutral publishers.

News Consumption Polarization on Social Media

Several recent studies have investigated the dissemination of news in social networks (Bhattacharya and Ram 2012), focusing on biases (Chakraborty et al. 2016), political news (An et al. 2014), and the characteristics of spreaders (Hu et al. 2012). More recently, the fragmentation of social media platforms following the acquisition of Twitter by Elon Musk in 2022 has spurred renewed interest in how partisan news consumption patterns persist and evolve across platforms such as Bluesky and Mastodon (Chuai and Zhao 2023; La Morgia et al. 2023; Xiao et al. 2024).

Traditionally, professional news organizations played a major role in spreading news by selectively presenting news stories to citizens (Shoemaker and Vos 2009). Accordingly, news media had a high impact on political issues and public opinions (Grosseclose and Milyo 2005; Chiang and Knight 2011). Several works have focused on understanding how and to what extent news media outlets can impact people and society, such as the White Helmets in Syria (Starbird et al. 2018) and the 2016 US presidential campaign (Rizoio et al. 2018; Ribeiro et al. 2019). Recent work has further shown that media outlet choice itself is a strong predictor of political belief formation, with algorithmic curation amplifying pre-existing partisan preferences (Guess et al. 2023; Huszár et al. 2022; Haroon et al. 2022).

By examining cross-ideological exposure through content and network analysis, Himmelboim, McCreery, and Smith (2013) showed that political talk on Twitter is highly partisan and users are unlikely to be exposed to cross-ideological content through their friendship network. Other studies also report similar findings such as users’ higher willingness to communicate with other like-minded social media users (Liu and Weber 2014). These echo chamber dynamics have been shown to accelerate the spread of misinformation, as low-consensus partisan content circulates rapidly within ideologically homogeneous clusters while rarely crossing community boundaries (Cinelli et al. 2021; Del Vicario et al. 2016; Vosoughi, Roy, and Aral 2018).

To understand the political bias in social media better, many researchers have studied political polarization on Twitter by analyzing different groups’ behavior. Conover et al. (2011) showed that Twitter users usually retweet users who have the same political ideology as themselves, making the retweet-

ing network structure highly partitioned into left- and right-leaning groups with limited connections between them. Similar partisan homophily patterns have been documented across different countries and political contexts, suggesting the phenomenon is not limited to US politics (An et al. 2023; Yarchi, Baden, and Kligler-Vilenchik 2021; Rauchfleisch and Kaiser 2020).

Previous work has mostly investigated news media political bias and the bias introduced in the content of the news by different methods such as crowdsourcing and machine learning (Budak, Goel, and Rao 2014; Gentzkow and Shapiro 2010; Babaei et al. 2018). Recently, a complementary approach was proposed by Babaei et al. (2018), in which the goal is to inject diversity in users’ information consumption by identifying high consensus yet informative news, based on using features such as the publishers’ political leaning.

Babaei et al. (2018) showed that high and low consensus posts are equally popular and cover broadly similar topics. However, their study did not investigate how low and high consensus news spread through social media, and their potential impacts on readers’ biased exposure, which are the main concerns of our paper. In this work, we first investigate how low and high consensus news spread through Twitter. Then we study the effect of spreading high consensus news through users with different political leanings on decreasing the disparity in users’ exposure. We show that seeding high consensus information with neutral individuals is the most effective way to achieve high spread with little disparity regarding political leaning.

LLM-based Annotation for Political and Social Science

A key bottleneck in studying consensus at scale has been the reliance on costly human annotation, typically via crowdsourcing platforms such as Amazon Mechanical Turk. Recent advances in large language models (LLMs) have opened a new avenue for scalable, low-cost annotation of political and social media content (Gilardi, Alizadeh, and Kubli 2023; Törnberg 2023). Gilardi, Alizadeh, and Kubli (2023) demonstrated that ChatGPT outperforms crowd workers on several political text annotation tasks including relevance, stance, topic, and framing classification, at a fraction of the cost. Similarly, Törnberg (2023) showed that GPT-4 surpasses both expert coders and crowd workers in annotating political Twitter

messages in a zero-shot setting.

These findings have motivated a growing body of work examining whether LLMs can serve as reliable annotators across a broader range of computational social science tasks (Ziems et al. 2024; Heseltine and Lasser 2024; Alizadeh et al. 2023). Ziems et al. (2024) evaluated LLMs on 24 CSS tasks and found that while performance is task-dependent, LLMs provide a strong zero-shot baseline that often matches supervised models trained on modest datasets. Alizadeh et al. (2023) further showed that open-source LLMs, when appropriately prompted, achieve competitive annotation quality on political text without the privacy concerns associated with proprietary API-based solutions.

Despite this progress, important questions remain about the consistency and potential biases of LLM annotators, particularly when applied to politically sensitive content (Mellon et al. 2024; Santurkar et al. 2023; Feng et al. 2023). Mellon et al. (2024) showed that LLM responses to political attitude questions correlate with human survey data but exhibit systematic skews depending on model choice and prompt framing. Santurkar et al. (2023) found that LLM opinions tend to reflect the perspectives of certain demographic groups more than others, raising concerns about representational bias in automated annotation pipelines.

In this work, we address these concerns by treating the LLM classifier as a *distributional* tool rather than an item-level ground truth: we validate that aggregate publisher-level consensus patterns produced by the classifier match those originally observed via AMT annotation, and we cross-check results across multiple model families to guard against single-model artifacts (Wang and Waseem 2021; Pangakis, Wolken, and Fasching 2023). This positions our contribution as a scalable extension of the original annotation methodology rather than a wholesale replacement of human judgment.

Information Propagation in Social Networks

The process of increasing information propagation and network diffusion by identifying and choosing the optimal set of individuals that utilize social influences to maximize adoption or reception of information in society has been studied widely (Goyal et al. 2013; Richardson and Domingos 2002; Kempe, Kleinberg, and Tardos 2003; Hartline, Mirrokni, and Sundararajan 2008). The effectiveness of these strategies is studied by Kempe, Kleinberg, and Tardos (2003) under different social contagion models such as Linear Threshold (LT) and Independent Cascade (IC) models. Recent work has extended classical influence maximization to account for temporal dynamics, network uncertainty, and the interplay between organic virality and algorithmic promotion (Chen et al. 2022; Banerjee, Jenamani, and Pratihari 2020; Li et al. 2018).

The goal of our paper is to propagate important and reliable news with less disparity amongst users with different political leanings. Two studies on fair influence maximization (Ali et al. 2019; Tsang et al. 2019b) are the most related to our work. However, their approaches may not be directly applicable to online networks such as Twitter. We show that in social media, the political leaning of the seeds can make a considerable bias in users’ exposure to news. In particular, we observe that high consensus news posted by neutral publishers has the lowest disparity for spreading among all users (liberal, conservative and neutral). More recent work on group-fair influence maximization has further explored the tension between reach maximization and equitable exposure across

demographic and ideological groups (Fish et al. 2019; Khajehnejad et al. 2022; Becker et al. 2022), reinforcing the importance of seed selection strategies that are aware of community structure. We use the fair influence maximization method proposed by Ali et al. (2019) as a baseline.

In the following sections, we explain our dataset and research design, and discuss our findings.

Dataset

In this work, we consider the dataset of 400 news tweets posted by 10 publishers collected in (Babaei et al. 2018) between 9th to 15th May, 2017. The dataset contains 80 low (20% of news with lowest consensus value) and 80 high consensus (20% of news with highest consensus value) news posts. Furthermore, we collected an additional 10 high and 10 low consensus recent news posts for each of COVID-19 and global warming, including 40 tweets in total.

To obtain the political leanings of the users who either retweeted a high or low consensus news or were exposed to it, for every high or low consensus news post in the dataset, we collected a random set of its 100 retweeters. Then for each retweeter, we collected a random set of his 100 followers. Finally, for each of these 1,616,000 users we collected their followees to compute their political leaning.

High and Low Consensus News Posts

Our news dataset consists of 10 news publishers with different political leanings varying from liberals to neutrals to conservatives: Slate, Salon, New York Times, CNN, AP, Reuters, Politico, Fox News, Drudge Report, and Breitbart News. From each publisher, 40 tweeted news posts are collected during the one week period of 9th to 15th May, 2017. For each news post, the authors in (Babaei et al. 2018) set up a survey in Amazon Mechanical Turk. They asked US AMT workers about their reaction to the post by selecting one out of three options: agreement, neutral, or disagreement. At the end of the experiment, they asked about AMT workers’ political leanings: liberal, conservative, or neutral.

Based on the above intuition, Babaei et al. (2018) proposed to capture the degree of consensus that a social media post is likely to have based on the distribution of the political leanings of the retweeters and repliers of a post, as follows.

$$\text{consensus} = 1 - \left| \frac{\#D_{\text{disagree}}}{\#D} - \frac{\#R_{\text{disagree}}}{\#R} \right|, \quad (1)$$

where $\#D_{\text{disagree}}$ and $\#R_{\text{disagree}}$ respectively denote the number of democrats and republicans who disagree with the post, while $\#D$ and $\#R$ are the total number of democrats and republicans. Authors also proposed a method to detect high and low consensus news automatically, which we use in our work to find an additional set of 40 high and low consensus tweets from the above mentioned publishers related to COVID-19 and global warming.

Note that high consensus news is distinct from low attention news. All stories used in our experiments discuss salient health, social, and political news – not lightweight gossip which may not generate much attention from or disagreement between users due to the chosen topics (see Section for more discussion about the newsworthiness of our news stories). Table 1 shows random sample news that are labeled as low consensus with conservative leaning, low consensus with liberal leaning, and high consensus news respectively.

Low Consensus News by Conservative Publishers
FoxNews: Far-left Democrats keep saving Republicans from themselves. Ainsley Earhardt: Trust Trump on Iran, he knows what's right.
FoxNews: Democrats own the chaos and racism at New Jersey anti-ICE riots. 11.5k Replies, 14.1k Likes
Breitbart: Illegal alien accused of brutal machete murder was released into US by Biden administration.
DrudgeReport: LIMBAUGH: TRUMP TROLLS DEMS with Comey firing. Violence and intimidation against Republicans becoming new normal.
Low Consensus News by Liberal Publishers
CNN: Trump insists US-Iran talks continue after Tehran suspended negotiations — critics call it capitulation.
CNN: James Comey's firing is a "grotesque abuse of power," CNN legal analyst Jeffrey Toobin says. 51 Retweets, 34 Likes
Salon: Trump can't sell his lies even in Versailles. UFC fight at the White House can't hide MAGA men's weakness.
Slate: Republicans used to support a watchdog role for the press. Not anymore. Stephen Miller was closer to realizing his scariest idea than we ever knew.
High Consensus News
Reuters: US and Iran reach agreement that includes opening of Strait of Hormuz.
AP: Trump signs peace agreement between United States and Iran at Versailles G7 summit.
Reuters: FDA approves new Alzheimer's treatment after successful Phase 3 trial results.
Politico: House passes resolution limiting Trump's war powers in Iran. 473 retweets, 39 replies, 573 likes
NYTimes: Supreme Court rules marijuana user cannot be barred from owning a gun.

Table 1: Samples of high and low consensus news posts. The first section shows low consensus news from conservative publishers, the second from liberal publishers, and the third shows high consensus news examples collected in 2026 using the LLM-based pipeline described in Section .

Figure 1(a) shows the number of high consensus (purple bars) and low consensus (orange bars) news posted by 10 publishers with various political leanings. Figure 1(b) shows the total number of high consensus and low consensus news posted by the same 10 publishers, grouped into liberal, conservative, and neutral categories. It also shows the total number of high consensus and low consensus news posted by all the 10 publishers. We can see that the total number of low consensus posts are considerably higher than the total number of high consensus posts.

As we discuss later, low consensus posts have a much smaller chance of being received by users with different political leanings, which leads to a more politically fragmented society. On the other hand, high consensus posts have a better chance of spreading through communities with various political leanings, and can be used to help break echo chambers.

Collecting Users' Political Leanings

For each post in our set of 80 high and 80 low consensus news posts, we collected a random set of its 100 retweeters. Then for each retweeter, we collected a random set of its 100 followers. Thus we have 1,616,000 twitter users. To infer a user's political leaning, one popular way is to apply NLP methods on the contexts that the user posts. However, tweets' context length is small, so we must look at users' neighbors. We inferred every user's political leaning, as a score between -1, +1, using the method of Kulshrestha et al. (2017), for which we needed to collect their followees. Inferring the political leaning of a given Twitter user u is based on the following steps – (i) generating two representative sets of

users who are known to have a democratic or republican bias, (ii) inferring the topical interests of u by looking at her followees, and (iii) examining how closely u 's interests match with the interests of the representative sets of democratic and republican users. Formally,

$$leaning(u) = cos_sim(I_u, I_D) - cos_sim(I_u, I_R), \quad (2)$$

where I_u is the interest vectors of user u , and I_D, I_R are normalized aggregate interest vector for the democrat seed set (I_D) and the republican seed set (I_R). Similarity between interest vectors are measured by cosine similarity.

For retweeters with certain political leaning, we calculated the expected fraction of their liberal, conservative, and neutral followers, as is shown in Table 2.

	Liberal	Conservative	Neutral
Liberal	0.76	0.04	0.2
Conservative	0.045	0.85	0.1
Neutral	0.3	0.27	0.43

Table 2: Expected fraction of liberal, conservative, and neutral followers of retweeters with various political leanings. Rows and columns correspond to retweeters and followers.

We also estimated the conditional probability that users with different political leanings retweet high consensus and low consensus news post from liberal, conservative, and neutral publishers (given that they retweet) in Table 3. It can be seen that users with a certain political leaning retweet low consensus posts from the publishers with the same political leaning with a very high probability. Interestingly, users retweet high consensus news posts from the publishers with the same political leaning with a smaller probability. On the other hand, there is a very small chance that users with a certain political leaning retweet low consensus posts from the publishers with different political leanings. For high consensus news this probability is larger.

		Retweeters		
		Liberal	Conservative	Neutral
Publishers	Liberal	H: 0.65 L: 0.85	H: 0.08 L: 0.04	H: 0.27 L: 0.11
		H: 0.12 L: 0.08	H: 0.68 L: 0.85	H: 0.2 L: 0.07
	Conservative	H: 0.34 L: 0.38	H: 0.33 L: 0.37	H: 0.33 L: 0.25
		H: 0.12 L: 0.08	H: 0.68 L: 0.85	H: 0.2 L: 0.07
	Neutral	H: 0.34 L: 0.38	H: 0.33 L: 0.37	H: 0.33 L: 0.25
		H: 0.12 L: 0.08	H: 0.68 L: 0.85	H: 0.2 L: 0.07

Table 3: Conditional probability of retweeting a high and low consensus news post (indicated H and L in the table) by users from various political leanings (given that they retweet). Rows and columns correspond to publishers and retweeters. For instance, in the first cell (first row and column), the probability that liberal users retweet high/low consensus news posts published by liberals publishers is 0.65/0.85.

LLM-based Consensus Detection and Temporal Validation

The original consensus labeling pipeline of Babaei et al. (2018) relies on Amazon Mechanical Turk workers to annotate each candidate post, eliciting reactions (agree / neutral /

disagree) from labelers with self-reported political leanings. While this yields high-quality ground truth, the per-post annotation cost imposes a hard ceiling on dataset size and prevents continuous monitoring of incoming posts over time (Gibaldi, Alizadeh, and Kubli 2023; Alizadeh et al. 2023).

Method. We replace the AMT annotation stage with a zero-shot LLM classifier. Given a post’s text, we prompt a locally-hosted instruction-tuned language model (Llama-3.1-8B, temperature = 0) to predict whether the post is HIGH or LOW consensus under the definition of Eq. (1): HIGH if the predicted liberal and conservative reaction rates are within δ of one another, LOW otherwise (Törnberg 2023; Ziems et al. 2024). The prompt enumerates the HIGH signals — factual reporting, scientific findings, official statements, neutral descriptions of events — and the LOW signals — partisan framing, ideologically loaded language, attacks on political opponents, op-ed framing, and topics that historically split on party lines. The model returns a one-word classification (HIGH or LOW); outputs that do not match either label are re-prompted at most once. Per-post inference requires approximately one second on a single consumer GPU, reducing the annotation cost from \$0.10–0.30 per post (AMT, multiple workers) to effectively zero, and enabling classification of arbitrarily large corpora without a fixed labeling budget (Heseltine and Lasser 2024; Pangakis, Wolken, and Fasching 2023).

Replication on the 2017 dataset. Figure 2 shows the result of applying the LLM classifier to the original 450 tweets from the 2017 dataset (50 per publisher, 9 publishers). The distribution of high and low consensus posts across publishers and leaning groups closely mirrors the pattern originally observed via AMT annotation in Babaei et al. (2018): neutral publishers (AP, Reuters) produce the highest share of high-consensus content, Salon and Slate the lowest, and the aggregate ordering Conservative < Liberal < Neutral is preserved. This confirms that the LLM pipeline reproduces the original findings at the distributional level, providing empirical justification for using it as a scalable substitute for crowd-sourced annotation.

Temporal validation on 2026 data. Figure 3 extends this analysis to a fresh corpus of ~1,000 political news items collected in 2026 across the same 10 publishers (~100 per publisher). The qualitative ordering is unchanged nearly a decade after the original study: neutral publishers continue to produce substantially more high-consensus content (61%) than conservative (45%) or liberal (42%) outlets. This out-of-time replication confirms that the editorial tendencies observed in the original study reflect stable, long-term publisher characteristics rather than artefacts of the 2017 news cycle (An et al. 2023), and demonstrates that the LLM-based pipeline can sustain the analysis pipeline beyond the constraints of the original AMT-labeled corpus.

The Gap between Proliferation of High and Low Consensus News

In this section, we investigate how high and low consensus news posts spread among users with various political leanings in Twitter. In particular, our goal is to answer the following key question:

How do individuals with certain political leaning (liberal, conservative, and neutral) get exposed to high and low consensus news posts?

Studying the above key question allows to understand the gap between proliferation of high and low consensus news, and develop strategies to decrease the polarization in the society by breaking the echo chambers that trap users. We start by investigating users’ behavior in retweeting high and low consensus news posts. Then, we discuss how the confirmation bias in retweeting behavior makes the echo chambers grow larger and promote social polarization.

Confirmation Bias in Retweeting Behavior

First, we study how users with different political leanings *share* high and low consensus news post. Specifically, we compare how users with different political leanings retweet low and high consensus news posts from publishers with different political perspectives.

Figures 4(a), 4(c), 4(e) show the distribution of the political leanings of all retweeters for one random low consensus and one random high consensus news posted by CNN (liberal publisher), FoxNews (conservative publisher), and Reuters (neutral publisher)⁴. Notice that the distribution of retweeters’ political leanings in Figure 4(a), 4(e) has more density in the right (liberal leaning) for the low consensus news. On the other hand, the distribution of retweeters’ political leanings in Figure 4(c) has considerably more density in the left (conservative leaning) for the low consensus news. Importantly, the distribution of retweeters’ political leanings for high consensus news (purple curve) is more symmetric in all the Figures. Moreover, the mean of the distribution for high consensus news is close to 0.

Next, we consider 80 low consensus and 80 high consensus news posted by the 10 publishers, ranging from liberals to neutrals to conservatives: Slate, Salon, New York Times, CNN, AP, Reuters, Politico, Fox News, Drudge Report, and Breitbart News. For each news post, we consider a set of its 100 retweeters chosen at random. Figures 4(b), 4(d), 4(f) show the distribution of the *expected* political leanings of retweeters of all the low consensus and high consensus news posted by liberal, conservative, and neutral publishers, respectively. Again, the distribution of retweeters’ political leanings for high consensus news (purple curve) is more symmetric, and is centered around 0 in all the Figures. In particular, the distribution of retweeters’ political leanings for high consensus news posted by neutral publishers has the most symmetric shape around 0.

We summarize our key observations as follows:

- I. Low consensus news posted by publishers with a specific political leaning (liberal/conservative) are mostly retweeted by users with similar political leanings (Figures 4(b), 4(d)).
- II. High consensus news posted by publishers with a specific political leaning (liberal/conservative) are retweeted by users with various political leanings (liberal/conservative/neutral) (Figures 4(b), 4(d)).
- III. While low and high consensus news posted by neutral publishers spread with lower disparity among users with different political leanings, high consensus news posted by

⁴The PDFs have been empirically estimated using kernel density estimation (Bowman and Azzalini 2004)

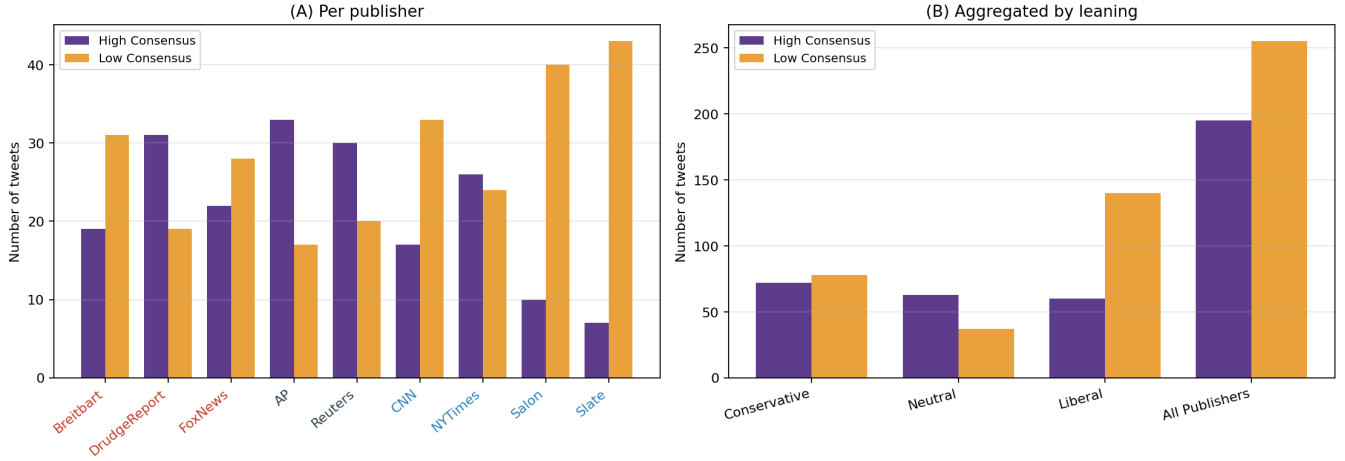


Figure 2: LLM-based consensus classification applied to the original 2017 tweet dataset (450 tweets, 9 publishers, 50 per publisher). (A) Per-publisher distribution of high and low consensus posts. (B) Aggregated by political leaning. The pattern closely reproduces that originally observed via AMT crowdsourcing in Babaei et al. (2018), confirming that the LLM classifier faithfully captures the same distributional signal at negligible annotation cost.

neutral publishers have the highest probability to be spread with minimum disparity among users (Figure 4(f)).

The Growth of Echo Chambers in Twitter

Next, we investigate how individuals with different political leanings get *exposed* to high and low consensus news posted by liberal, conservative, and neutral publishers.

Figure 5(a) depicts the distribution of political leanings for followers (level 1) of liberal, conservative, and neutral publishers. We observe that users with conservative or liberal leanings are mostly exposed to news posted by publishers with the same political leaning (the chamber effect). Therefore, the distribution of political leaning for followers of liberal and conservative publishers are skewed to the left and right, respectively. Nevertheless, followers of conservative publishers has a more skewed distribution. This is resulted from the fact that the conservative community is denser, and has fewer connections to liberals and neutrals in Twitter (*c.f.* Table 2). On the other hand, the distribution of political leanings for neutral users is very symmetric and is centered at 0. Hence, neutral users get similar exposure to liberal and conservative view points.

Figure 5(b) shows the distribution of political leanings for followers of retweeters (level 2) of liberal, conservative, and neutral publishers. We observe that while the distribution of political leanings for followers of retweeters of neutral publishers is symmetric and centered around 0, the distribution of political leanings for followers of retweeters of liberal or conservative publishers are extremely skewed. As expected, followers of retweeters of conservative publishers have a more skewed distribution. Interestingly, the skewness of the distributions for followers of retweeters (level 2) is much larger compared to the skewness of distributions for followers of publishers (level 1). This means that echo chambers in level 2 are larger than those in level 1. Our experiments show that as we get farther away from the publishers, echo chambers grow even larger (Figure 10).

We summarize our key observations as follows:

- I. Conservatives and liberals get a biased exposure to the news posted in Twitter, while neutrals get similar exposure to liberal and conservative view points (Figure 5(a)).
- II. Users who get the news from retweeters get a significantly more biased exposure, compare to users who get the news from publishers. In other words, as we get farther away from the news publishers, the echo chambers grow larger (Figure 5(b)).

Breaking Echo Chambers

To break echo chambers, we aim for all individuals to get similar exposure to news stories. Our proposed strategy that we hope to break the chamber effect is based on the three key observations discussed earlier: (1) while low consensus news are more likely to proliferate amongst the users with a particular political leaning, high consensus news has a much higher chance of spreading among users with different political leanings; (2) high consensus news posted by neutral publishers has the lowest disparity for spreading among liberal and conservative users; and (3) as users get farther away from publishers, they get a more biased exposure to news. Based on the above observations, we conjecture:

High consensus news posted by neutral users may help to break echo chambers.

High consensus news posted by neutral users achieve high spread with little disparity regarding political leaning. We confirm our conjecture and show the effectiveness of our proposed strategy through an extensive set of experiments later in the paper.

In the following section, we first formulate the problem of information diffusion in social networks. Then, we discuss the problem of finding a near-optimal set of neutral users to seed spreading high consensus news and break the echo chamber.

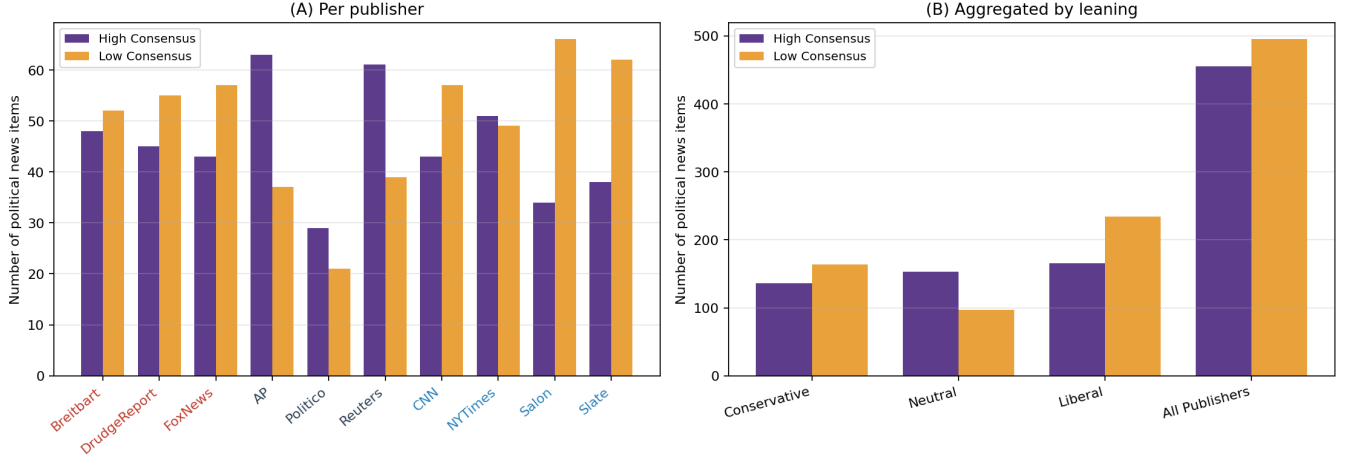


Figure 3: LLM-based consensus classification applied to a fresh corpus of $\sim 1,000$ political news items collected in 2026 across the same 10 publishers (~ 100 per publisher). (A) Per-publisher distribution of high and low consensus posts. (B) Aggregated by political leaning. The core pattern is preserved: neutral publishers produce the highest fraction of high-consensus content (61%), while conservative (45%) and liberal (42%) outlets both skew towards low-consensus political framing, providing strong out-of-time empirical corroboration of the paper’s central claim.

Problem Formulation: Information Diffusion

We start by formulating the information diffusion problem to model the spread of news among individuals with various political leanings in Twitter. We simulate the proliferation of news by assuming that each user can be a publisher. Then we select a set of users and involve them to post news. We represent Twitter by a directed graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of directed edges between the nodes. The nodes in the network are partitioned into three disjoint groups $\mathcal{V} = \{\mathcal{V}_d, \mathcal{V}_r, \mathcal{V}_n\}$, where $\mathcal{V}_d, \mathcal{V}_r, \mathcal{V}_n$ represent users with liberal, conservative, and neutral leanings, respectively. A directed edge (v, u) exists if user v follows user u . When users post tweets, their followers can retweet and spread the tweets in the network. To model spread of information, e.g. news or tweets in Twitter, two well-known classical diffusion models are introduced in the literature (Pastor-Satorras and Vespignani 2001): (1) Independent Cascade model (IC) and (2) Linear Threshold (LT) model. In this work, we consider the IC model.

Independent Cascade model (IC)

In the IC model, information propagates through every edge (v, w) with probability p_{vw} . We have a set of discrete time steps which we denote with $t = \{0, 1, 2, \dots\}$. At $t = 0$, the initial seed set $S \subseteq \mathcal{V}$ is activated. At every time step $t > 0$, a node $v \in V$ which was activated at time $t - 1$ can activate its inactivated neighbors w with probability p_{vw} . The model assumes that once a node is activated, it stays active throughout the whole process and each node has only one chance to activate its neighbors. The described process stops at time $t > 0$ if no new node gets activated at this time. We note that the IC model is a stochastic process, in which a node u can influence its neighbors w based on the Bernoulli distribution with success probability p_{uw} . A possible outcome of the process can be denoted via a set of timestamps $\{t_v \geq 0 : v \in \mathcal{V}\}$, where t_v represents the time at which a node $v \in V$ is activated.

Information Diffusion with Low Disparity

Our goal is to find the smallest seed set of users that when post a tweet, it spreads through at least a fraction $Q_p \in [0, 1]$ of liberals ($\mathcal{V}_l$), conservatives ($\mathcal{V}_c$), and neutrals ($\mathcal{V}_n$) in Twitter, where $p \in \{l, c, n\}$. We formulate the problem as follows:

$$\min_{S \subseteq \mathcal{V}} |S| \quad \text{subject to} \quad (3)$$

$$\sum_{p \in \{l, c, n\}} \min(f_p(S), Q_p \cdot |\mathcal{V}_p|) \geq \sum_{p \in \{l, c, n\}} Q_p,$$

where $f_l(\cdot), f_c(\cdot), f_n(\cdot)$ determine the total number users among liberals ($\mathcal{V}_l$), conservatives ($\mathcal{V}_c$), and neutrals ($\mathcal{V}_n$) that are activated as a result of selecting the seed set S . We call $\mathcal{V}_p$ “saturated” by S when $\min(f_p(S), Q_p \cdot |\mathcal{V}_p|) = Q_p \cdot |\mathcal{V}_p|$. When a certain fraction Q_p of individuals with a particular political leaning p are exposed to a news (activated), any new activated individual with political leaning p cannot further improve the utility. This will give individuals with different political leanings a higher chance of being exposed to the news.

We note that the utility function, i.e., $f_p : 2^{\mathcal{V}_p} \rightarrow \mathbb{Z}^+$, is a non-negative, monotone, submodular set function (Kempe, Kleinberg, and Tardos 2003). The submodularity is an intuitive notion of diminishing returns, stating that for any sets $A \subseteq A' \subseteq V$ and any node $a \in V \setminus A'$, it holds that:

$$f(A \cup \{a\}) - f(A) \geq f(A' \cup \{a\}) - f(A').$$

Although problem (3) is NP-hard in general (Wolsey 1982), for maximizing a submodular function the following greedy algorithm provides a logarithmic approximation guarantee. The greedy algorithm starts from an empty set, add a new node to the set which provides the maximal marginal gain in terms of utility, and stops whenever the desired Q_p fraction of individuals with political leaning p are activated.

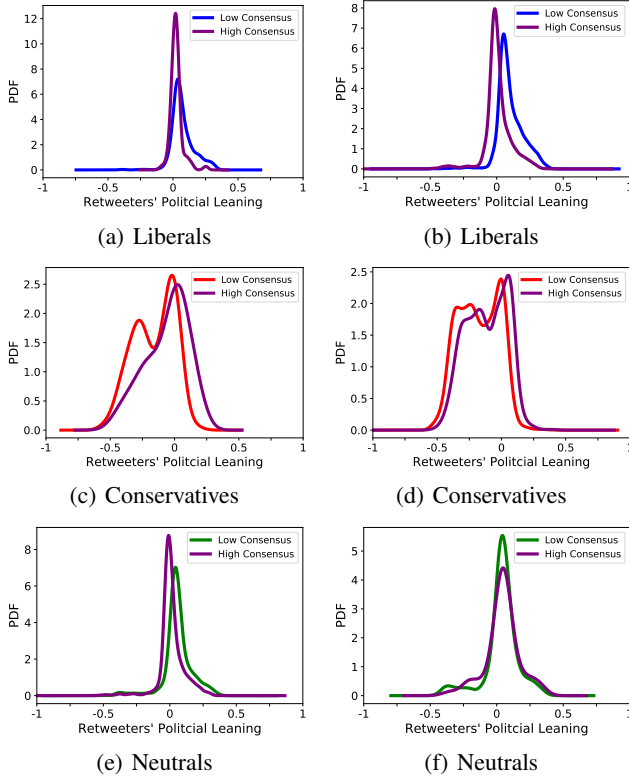


Figure 4: Distribution of retweeters’ political leanings for low and high consensus news posted by publishers with different political perspectives. Distribution of political leanings for a random high and a random low consensus news posted by (A) liberal, (C) conservative, and (E) neutral publishers. Distribution of average political leanings for 100 high and low consensus news posted by (B) liberal, (D) conservative, and (F) neutral publishers. Distribution of political leanings for high consensus news (purple) is more symmetric and centered around 0.

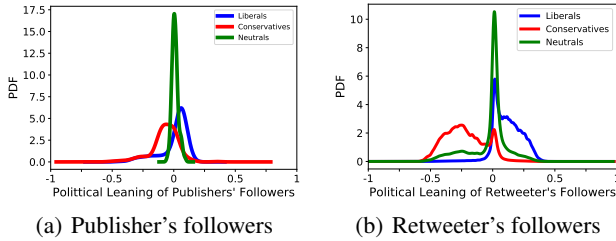


Figure 5: Distribution of political leanings for (A) followers of liberal, conservative, and neutral publishers, and (B) followers of retweeters of liberal, conservative, and neutral publishers. As we get farther away from the publishers, the distribution of liberal and conservative followers becomes significantly more skewed (echo chambers grow larger).

Spreading through Neutrals to Break the Chambers

In Problem (3), the fraction Q_p can be arbitrary for individuals with different political leanings. However, to break the

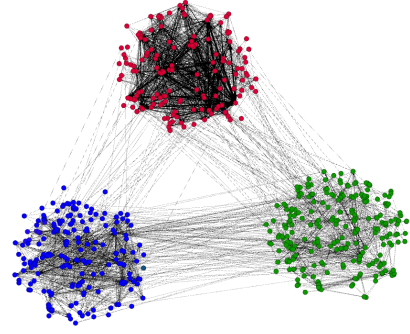


Figure 6: The sample graph from the real Twitter data set collected in 2009. Blue, red, and green nodes indicate users with liberal, conservative, and neutral political leanings.

echo chambers we wish individuals with different political leanings to get a similar exposure to various news. In other words, we assume similar values for Q_l, Q_c, Q_n . Moreover, the news posted by individuals with neutral leanings have a higher chance of spreading among individuals with liberal and conservative political leanings. Therefore, to break the echo chambers we aim at finding the smallest subset $S \subseteq \mathcal{V}_n$ that when post a news, at least a fraction Q_p of individuals with political leaning p get exposed to the news. Formally, we have

$$\min_{S \subseteq \mathcal{V}_n} |S| \quad \text{subject to} \quad (4)$$

$$\sum_{p \in \{l, c, n\}} \min(f_p(S), Q_p \cdot |\mathcal{V}_p|) \geq \sum_{p \in \{l, c, n\}} Q_p.$$

Experimental Results

In this section we investigate the effect of spreading high consensus news posted by neutral users among individuals with conservative, liberal, and neutral leanings in Twitter. In particular, we show that our proposed strategy is very effective in spreading information among individuals with various political leanings and lowering societal polarization for news consumption. We first describe our instance of Twitter network. We then explain our experimental setup, and present our findings.

Twitter Network. Our network is collected from Twitter in September 2009 (Babaei et al. 2016; Cha et al. 2010), and includes: 52 million user profiles, 1.9 billion directed follow links among the users, and 1.7 billion public tweets posted by the users. In order to obtain a static network, we consider the tweets published on July 1, 2009, and filter out users that did not tweet before July 1. After this filtering, we have 70,000 active users. We then extract the strongest connected community, including 69,687 users and 2,907,026 link between them, yielding 19162, 3449, 47076 nodes with liberal, conservative, and neutral leanings, respectively. The average degree of network is 41.5. Figure 6 shows an induced random sample from our final Twitter network.

Sampled Twitter Network. We also created a smaller network by sampling 10% of nodes uniformly at random from our original Twitter network, and connecting the users if they

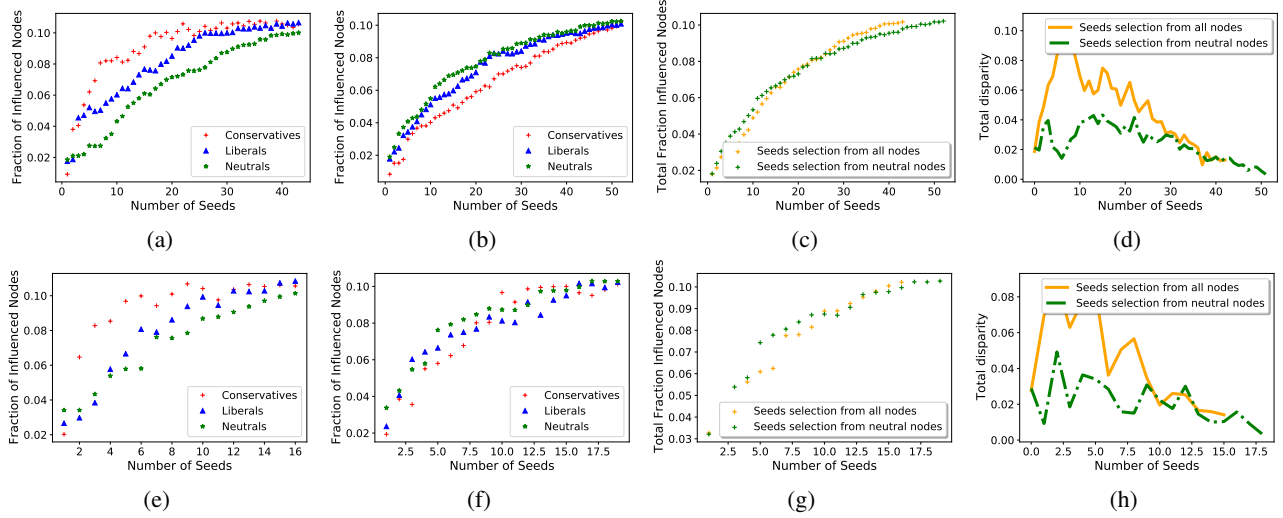


Figure 7: Fraction of individuals with liberal, conservative, and neutral political leanings who are exposed to a high consensus news. Top row shows the result on our Twitter network, and the bottom row shows the result on the smaller sampled Twitter network. (A), (E) show the fraction of exposed individuals when the seeds are selected from the entire network by solving Problem (3). (B), (F) show the fraction of exposed individuals when the seeds are selected from neutral users by solving Problem (4). (C), (G) compare the fraction of exposed individuals when the initial seed is selected from the entire network vs. neutrals. (D), (H) compare the disparity of diffusion when the initial seed set is selected from the entire network vs. neutrals.

have a connection in the original network. The strongest connected community includes 3,753 users and 6,993 connections with average degree of 1.83. Our sampled Twitter network includes 812 liberals, 186 conservatives, and 2,755 neutrals. Note that the structure of the original Twitter network is very different than the sampled Twitter network. In particular, the sampled Twitter network is significantly sparser than the original Twitter network.

Experimental Setup. For a pair of users $u \in \mathcal{V}_i$ and $w \in \mathcal{V}_j$, we calculate the success probability of activation p_{uw} as the expected fraction of users with political leaning j who retweeted news posted by users with political leaning i . The retweeting probabilities are listed in Table 3.

We apply the greedy algorithm to find a near optimal subset of users that can spread a news over a certain fraction $Q_l = Q_c = Q_n = 0.1$ of liberals ($\mathcal{V}_l$), conservatives ($\mathcal{V}_c$), and neutrals ($\mathcal{V}_n$) in the Twitter network. To evaluate the utility function $f_p(\cdot)$ in Problem 3 and Problem 4, we estimate it by using Monte Carlo sampling (Hastings 1970). We used 200 samples for this estimation, which yielded a stable estimation of the utility function.

Note that using equal values for Q_l, Q_c, Q_n in Problem (3), we retrieve the fair influence maximization formulation proposed by (Ali et al. 2019). In our experiments, we compare our proposed strategy to fair influence maximization.

Neutrals Can Break Echo Chambers

In our first set of experiments, we apply the greedy algorithm to Problem (3) and Problem (4) to find the initial set of users to spread news in Twitter. Figure 7 compares the fraction of individuals with liberal, conservative, and neutral political leanings who got exposed to a high consensus news spread through an initial seed set obtained by solving Problem (3)

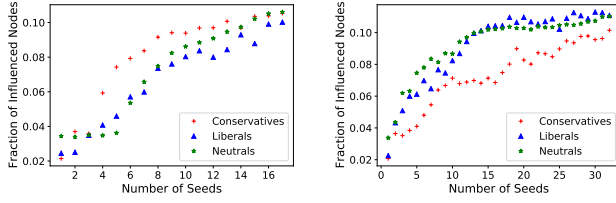
vs. Problem (4). The goal is to expose $Q_p = 10\%$ of individuals with liberal, conservative, and neutral leanings to the news. The top row shows the result on our original Twitter network, and the bottom row shows the result on the smaller sampled Twitter network. Note that the sampled network is much sparser than the original Twitter network.

Figures 7(a), 7(e) show the fraction of exposed individuals when the seeds are selected from the entire network by solving Problem (3). Figures 7(b), 7(f) show the fraction of exposed individuals when the seeds are selected from the users with neutral leanings by solving Problem (4). We note that as more individuals are added to the initial seed set by the greedy algorithm, the disparity in the number of exposed users with different political leanings is much smaller in Figures 7(b), 7(f) compared to Figures 7(a), 7(e). This clearly confirms the effectiveness of our proposed strategy in breaking the echo chambers.

We note that if we do not take into account the different pattern of diffusion among users of various political leanings, the neutral users may not be the ones that can maximize the spread of information. Figure 8(a), 8(b) compare the fraction of users who were exposed to a high-consensus news, when the seeds are selected from the entire network vs. only the neutral users. Here, we assume that all the users spread the news with the same probability of 0.1 irrespective of their political leanings. We see that the news posted by neutral publishers has more disparity and reaches a smaller number of users.

Neutrals Can Spread News Widely

Figures 7(c), 7(g) compare the fraction of exposed individuals when the initial seed is selected from the entire network vs. neutrals. There are two interesting observations: the initial seeds selected from neutral users can spread the news even more than users selected from the entire network. Moreover,



(a) Seeds with various leanings (b) Seeds with neutral leanings

Figure 8: Fraction of users who were exposed to a high-consensus news story, when all users propagate the news with the same probability of 0.1, irrespective of their political leanings in Twitter. (A) shows the result when seeds are selected from the entire network, (B) shows the result when seed are selected only from the neutral users.

		Retweeters			
Publishers	Liberal	Liberal	Conservative	Neutral	Sum
		H: 76 L: 104	H: 9 L: 5	H: 32 L: 15	H:117 L:124
	Conservative	H: 9 L: 10	H: 58 L: 94	H: 18 L: 9	H:87 L:113
		H: 45 L: 49	H: 43 L: 47	H: 43 L: 32	H:131 L:128
	Neutral				

Table 4: Average number of retweeting a high and low consensus news post (indicated H and L in the table) by users from various political leanings. Rows and columns correspond to publishers and retweeters. For instance, in the first cell (first row and column), liberal users on average retweets high/low consensus news posts published by liberals publishers 76 times. We note that news posted by neutrals is retweeted by an even larger number of users, compared to news posted by liberal or conservative publishers.

as we continue the selection process, selected neutral seeds can spread the news as well as the seed set selected from the entire network. This interesting observation confirms the power of neutral users in spreading news in Twitter. As Table 4 depicts, the average number of retweeting of a tweet posted by liberal, conservative, and neutrals are almost equal. There is a interesting observation. High consensus tweets posted by neutrals are retweeted with many democrats and republicans in addition to neutrals. Interestingly, news posted by neutrals is retweeted by an even larger number of users compared to news posted by liberal or conservative publishers.

Figure 9(a) shows the number of users selected with liberal, conservative, and neutral leanings for varying number of seeds selected greedily to solve Problem (3). Figure 9(a) shows the result on the Twitter network, and Figure 9(b) shows the result on the sampled Twitter network. We see that in the set of seeds greedily selected from the entire network, the majority of the users have neutral leanings. This further shows that neutral users are highly effective in spreading information in Twitter. This is consistent with our initial observation, that the news posted by neutrals has a higher probability of spreading among users with different political leanings.

Neutrals Spread News with Low Disparity

Figures 7(d), 7(h) compare the total disparity of diffusion when the initial seed set is selected from the entire network vs. neutrals. We define the total disparity as the sum of all

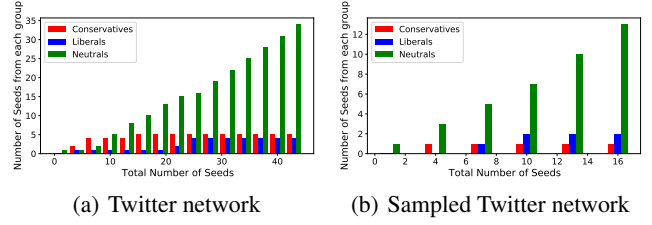


Figure 9: Number of users selected with liberal, conservative, and neutral leanings for varying number of seeds selected greedily to solve Problem (3). Figure shows the results on (A) the Twitter network, and (B) the smaller sampled Twitter network.

disparity (differences) between exposure for each pair of political leanings. Formally, we have: Total disparity=

$$\left| \frac{f_l(S)}{\mathcal{V}_l} - \frac{f_c(S)}{\mathcal{V}_c} \right| + \left| \frac{f_l(S)}{\mathcal{V}_l} - \frac{f_n(S)}{\mathcal{V}_n} \right| + \left| \frac{f_c(S)}{\mathcal{V}_c} - \frac{f_n(S)}{\mathcal{V}_n} \right|.$$

We observe that the total disparity is much smaller when the initial seed set is selected from users with neutral political leanings (Problem (3)) compared to the case when the initial seed set is selected from the entire network (Problem (4)). The difference is larger when the size of the initial seed set is smaller.

Echo Chambers Grow Larger over Time

Figure 10 shows the fraction of individuals with various political leanings who got exposed to a high consensus news during the diffusion process (IC), for varying number of seeds. More precisely, for a given seed set information diffusion proceeds in discrete time steps $t = \{0, 1, 2, \dots\}$. Figure 10 compares the fraction of users with various political leanings who received the news in the first time-step, $t = 1$, and second time-step $t = 2$ in our original Twitter network. Figure 10(a) shows the result when the seeds are selected from the entire network, by solving Problem (3). Figure 10(b) shows the result when the seeds are selected from the users with neutral political leanings, by solving Problem (4). It can be seen that when seeds are selected from the entire network, the disparity becomes larger as the diffusion process continues. On the other hand, the disparity is much smaller when seeds are selected from neutral users.

The above result confirms our observation that the echo chambers grow larger as the diffusion continues over time. In other words, when the seeds are selected from the entire network, as the we get farther away from the initial set of seeds, the disparity in the number of users with different political leanings who are exposed to the news becomes larger. On the other hand, when diffusion is originated from neutral seeds, users with different political leanings get exposed to the information at the same time. This is crucial while spreading time-critical information, such as health-related information or emergency warnings, in the network.

Discussion

Since we propose increasing exposure to high consensus news, we would like to check that such stories carry important information for public discourse. Babaei et al. (2018) compared low and high consensus posts on social media by empirically analyzing their properties. They showed that both

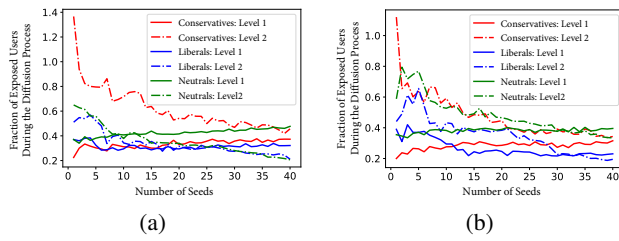


Figure 10: Fraction of users who are exposed to the news from different groups in first and second time step of propagation process in the Twitter network. (A) Shows the result when seed are selected from the entire network (Problem 3), (B) shows the result when seed are selected from neutral users (Problem 4).

types of posts are equally popular and cover similar topics. We checked this by analyzing 400 randomly selected posts including examples of high and low consensus news, along with their sources, number of retweets, replies, and likes. We highlight the following observations:

- I. For both types of news posts, a variety of news sources exists across the ideological spectrum. Figure 1 also shows several publishers with different political leaning that posts both types of high and low consensus news.
- II. On average, high and low consensus tweets are retweeted 158 and 177 times respectively. On average high consensus tweets are liked 532 times, whereas, low consensus news are liked 488 times. Thus, high and low consensus news stories have similar popularity.
- III. Galtung and Ruge (1965) introduce newsworthiness theory in which they propose several news factors such as frequency, meaningfulness, continuity, etc. Eilders (2006) showed that these factors impact news' worthiness. Weber (2014) proposes the following hypothesis: "The news factors of a news item influence the level of participation in commenting in an article's comments section". Weber also noted several other factors, such as having a high social impact or being controversial, that may attract more comments as participation (Weber 2012). Weber emphasized that if a news story attracts more comments, then it has higher worthiness. Here we can consider the number of replies as participating comments. On average, high consensus and low consensus news stories received 100 and 114 replies (comments), respectively, suggesting that both types of news have similar worthiness.

In summary, we observe that high and low consensus news are similar along multiple dimensions, including variety of news source, popularity, topic covering, and worthiness.

Conclusion

In this work, we studied the diffusion of news in Twitter. We investigated how users with various political leanings (liberals, conservatives and neutrals) get exposed to low and high consensus news posted by different publishers (e.g. CNN, FoxNews, etc.). We found that (1) while low consensus news stories are more likely to proliferate amongst users with a particular political leaning, high consensus news has a much higher chance of spreading among users with different political leanings and can reach a greater total number of users; (2) high consensus news posted by neutral publishers has the

lowest disparity for spreading among liberal and conservative users and reaches the greatest total number of users; and (3) as users get farther away from the publishers, they get a more biased exposure to the news.

Based on the above observations, we studied the effect of spreading high consensus news through neutral users on decreasing the disparity in users' exposure. Our extensive simulation experiments on Twitter showed that our proposed strategy can be highly effective in spreading information while decreasing the disparity of information across users with differing views.

While some of our results may appear unsurprising, we believe we are the first to empirically verify, and theoretically justify, that seeding high consensus news stories with neutral publishers is superior to other strategies (such as seeding with several publishers from each political leaning, particularly for a low number of initial seeds, see Fig 5 C) in reaching a large number of users, who are also diverse across their views. We hope our findings may be helpful for spreading newsworthy stories broadly and equitably, breaking echo chambers, reducing fragmentation in online social media, lowering segregation in consumption of important news, and help lead to a healthier society.

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References

- Ali, J.; Babaei, M.; Chakraborty, A.; Mirzasoleiman, B.; Gummadi, K. P.; and Singla, A. 2019. On the Fairness of Time-Critical Influence Maximization in Social Networks. *ArXiv abs/1905.06618*.
- Ali, J.; Babaei, M.; Chakraborty, A.; Mirzasoleiman, B.; Gummadi, K. P.; and Singla, A. 2022. On the Fairness of Time-Critical Influence Maximization in Social Networks. *IEEE Transactions on Knowledge and Data Engineering* 34(9): 4350–4363.
- Alizadeh, M.; Kubli, M.; Samei, Z.; Dehghani, S.; Bermeo, J. D.; Korobeynikova, M.; and Gilardi, F. 2023. Open-source LLMs for text annotation: A practical guide for model setting and fine-tuning. *arXiv preprint arXiv:2307.02179*.
- Altay, S.; Berriche, M.; and Acerbi, A. 2023. Misinformation on misinformation: Conceptual and methodological challenges. *Social Media + Society* 9(1).
- An, J.; Kwak, H.; Mejova, Y.; de Francisci Morales, G.; and Bonchi, F. 2023. Measuring political homophily on Twitter. *Social Networks* 72: 84–94.
- An, J.; Quercia, D.; Cha, M.; Gummadi, K.; and Crowcroft, J. 2014. Sharing political news: the balancing act of intimacy and socialization in selective exposure. *EPJ Data Science* 3(1): 12.
- Babaei, M.; Grabowicz, P.; Valera, I.; Gummadi, K. P.; and Gomez-Rodriguez, M. 2016. On the efficiency of the information networks in social media. In *Proceedings of the Ninth ACM International Conference on Web Search and Data Mining*, 83–92.

- Babaei, M.; Kulshrestha, J.; Chakraborty, A.; Benevenuto, F.; Gummadi, K. P.; and Weller, A. 2018. Purple feed: Identifying high consensus news posts on social media. In *Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics & Society (AIES)*.
- Bail, C. A. 2014. *Terrified: How anti-Muslim fringe organizations became mainstream*. Princeton University Press.
- Bail, C. A.; Argyle, L. P.; Brown, T. W.; Bumpus, J. P.; Chen, H.; Hunzaker, M. F.; Lee, J.; Mann, M.; Merhout, F.; and Volfovsky, A. 2018. Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences* 115(37): 9216–9221.
- Bakshy, E.; Messing, S.; and Adamic, L. 2015a. Exposure to ideologically diverse news and opinion on Facebook. *Science* .
- Bakshy, E.; Messing, S.; and Adamic, L. A. 2015b. Exposure to ideologically diverse news and opinion on Facebook. *Science* 348(6239): 1130–1132.
- Balayn, A.; Yang, J.; Szlávi, G.; and Bozzon, A. 2023. Searching for Political Perspectives: The Role of Search Engines in Political Information Seeking. *Proceedings of the ACM Web Conference* .
- Banerjee, S.; Jenamani, M.; and Pratihari, D. K. 2020. A survey on influence maximization in a social network. *Knowledge and Information Systems* 62(9): 3417–3455.
- Becker, R.; D’Angelo, G.; Ghobadi, S.; and Gilbert, H. 2022. Fairness in Influence Maximization through Randomization. In *Journal of Artificial Intelligence Research*, volume 73, 1251–1283. AI Access Foundation.
- Bhattacharya, D.; and Ram, S. 2012. Sharing news articles using 140 characters: A diffusion analysis on Twitter. In *2012 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*, 966–971. IEEE.
- Bowman, A. W.; and Azzalini, A. 2004. *Applied smoothing techniques for data analysis*. Clarendon Press.
- Bozdog, E. 2013. Bias in algorithmic filtering and personalization. *Ethics and Information Technology* 15(3): 209–227. ISSN 1572-8439. doi:10.1007/s10676-013-9321-6. URL https://doi.org/10.1007/s10676-013-9321-6.
- Budak, C.; Goel, S.; and Rao, J. M. 2014. Fair and Balanced? Quantifying Media Bias Through Crowdsourced Content Analysis. <http://dx.doi.org/10.2139/ssrn.2526461>.
- Cha, M.; Haddadi, H.; Benevenuto, F.; and Gummadi, K. P. 2010. Measuring user influence in Twitter: the million follower fallacy. In *Proceedings of AAAI Conference on Weblogs and Social Media (ICWSM’2010)*.
- Chakraborty, A.; Ali, M.; Ghosh, S.; Ganguly, N.; and Gummadi, K. P. 2017. On quantifying knowledge segregation in society. *arXiv preprint arXiv:1708.00670* .
- Chakraborty, A.; Ghosh, S.; Ganguly, N.; and Gummadi, K. P. 2016. Dissemination biases of social media channels: On the topical coverage of socially shared news. In *Tenth International AAAI Conference on Web and Social Media*.
- Chen, L.; Babaei, M.; Grabowicz, P.; and Gummadi, K. P. 2022. Temporal Dynamics of Online Petitions: The Role of Attention and Exposure in Collective Action. *Proceedings of the ACM on Human-Computer Interaction* 6(CSCW2).
- Chiang, C.-F.; and Knight, B. 2011. Media bias and influence: Evidence from newspaper endorsements. *The Review of Economic Studies* 78(3): 795–820.
- Chuai, Y.; and Zhao, J. 2023. Anger Makes Fake News Viral Online. In *Frontiers in Physics*, volume 11. Frontiers Media.
- Cinelli, M.; De Francisci Morales, G.; Galeazzi, A.; Quattrociocchi, W.; and Starnini, M. 2021. The echo chamber effect on social media. *Proceedings of the National Academy of Sciences* 118(9).
- Conover, M.; Ratkiewicz, J.; Francisco, M.; Gonçalves, B.; Menczer, F.; and Flammini, A. 2011. Political Polarization on Twitter. In *Proc. AAAI ICWSM*.
- Del Vicario, M.; Bessi, A.; Zollo, F.; Petroni, F.; Scala, A.; Caldarelli, G.; Stanley, H. E.; and Quattrociocchi, W. 2016. The spreading of misinformation online. *Proceedings of the National Academy of Sciences* 113(3): 554–559.
- Eady, G.; Nagler, J.; Guess, A.; Zilinsky, J.; and Tucker, J. A. 2019. How Many People Live in Political Bubbles on Social Media? Evidence From Linked Survey and Twitter Data. *SAGE Open* 9(1).
- Eilders, C. 2006. News factors and news decisions. Theoretical and methodological advances in Germany. *Communications* 31(1): 5–24.
- Feng, S.; Park, C. Y.; Liu, Y.; and Tsvetkov, Y. 2023. From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models 11737–11762.
- Fish, B.; Bashardoust, A.; Boyd, D.; Friedler, S.; Scheidegger, C.; and Venkatasubramanian, S. 2019. Gaps in Information Access in Social Networks. In *Proceedings of The Web Conference 2019*, 480–490. ACM.
- Flaxman, S.; Goel, S.; and Rao, J. M. 2016. Filter Bubbles, Echo Chambers, and Online News Consumption. *Public Opinion Quarterly* 80(S1): 298–320. doi:10.1093/poq/nfw006. URL +http://dx.doi.org/10.1093/poq/nfw006.
- Galtung, J.; and Ruge, M. H. 1965. The structure of foreign news: The presentation of the Congo, Cuba and Cyprus crises in four Norwegian newspapers. *Journal of peace research* 2(1): 64–90.
- Gentzkow, M.; and Shapiro, J. M. 2010. What Drives Media Slant? Evidence From U.S. Daily Newspapers. *Econometrica* 78: 35–71.
- Gilardi, F.; Alizadeh, M.; and Kubli, M. 2023. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences* 120(30).
- Goyal, A.; Bonchi, F.; Lakshmanan, L. V.; and Venkatasubramanian, S. 2013. On minimizing budget and time in influence propagation over social networks. *Social network analysis and mining* 3(2): 179–192.
- Groseclose, T.; and Milyo, J. 2005. A measure of media bias. *The Quarterly Journal of Economics* 120: 1191–1237.
- Gruzd, A.; and Mai, P. 2020. Going viral: How a single tweet spawned a COVID-19 conspiracy theory on Twitter. *Big Data & Society* 7(2): 2053951720938405.
- Guess, A. M.; Malhotra, N.; Pan, J.; Barberá, P.; Allcott, H.; Brown, T.; Crespo-Tenorio, A.; Dimmery, D.; Freelon, D.; Gentzkow, M.; et al. 2023. How do social media feed algorithms affect attitudes and behavior in an election campaign?

In *Science*, volume 381, 398–404. American Association for the Advancement of Science.

Harooun, M.; Chhabra, A.; Liu, X.; Mohapatra, P.; Shafiq, Z.; and Wojcieszak, M. 2022. YouTube Recommendations and Effects on Sharing Across Political Channels: Evidence from Two Experiments. In *Proceedings of the ACM on Human-Computer Interaction*, volume 6. ACM.

Hartline, J.; Mirrokni, V.; and Sundararajan, M. 2008. Optimal marketing strategies over social networks. In *Proceedings of the 17th international conference on World Wide Web*, 189–198.

Hastings, W. K. 1970. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika* 57(1): 97–109. ISSN 0006-3444. doi:10.1093/biomet/57.1.97. URL <https://doi.org/10.1093/biomet/57.1.97>.

Heseltine, M.; and Lasser, J. 2024. Large language models outperform expert coders and annotators for the annotation of political content. *arXiv preprint arXiv:2404.02986*.

Himelboim, I.; McCreery, S.; and Smith, M. 2013. Birds of a feather tweet together: Integrating network and content analyses to examine cross-ideology exposure on Twitter. *Journal of Computer-Mediated Communication* 18(2): 40–60.

Hu, M.; Liu, S.; Wei, F.; Wu, Y.; Stasko, J.; and Ma, K.-L. 2012. Breaking news on twitter. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2751–2754. ACM.

Huszár, F.; Ktena, S. I.; O’Brien, C.; Belli, L.; Schlaikjer, A.; and Hardt, M. 2022. Algorithmic amplification of politics on Twitter. In *Proceedings of the National Academy of Sciences*, volume 119. National Academy of Sciences.

Kempe, D.; Kleinberg, J.; and Tardos, É. 2003. Maximizing the spread of influence through a social network. In *KDD*.

Khajehnejad, M.; Khajehnejad, A.; Babaei, M.; Gummadi, K. P.; Weller, A.; and Mirzasoleiman, B. 2022. CrossWalk: Fairness-Enhanced Node Representation Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, 11963–11970. AAAI Press.

Kulshrestha, J.; Eslami, M.; Messias, J.; Zafar, M. B.; Ghosh, S.; Gummadi, K. P.; and Karahalios, K. 2017. Quantifying Search Bias: Investigating Sources of Bias for Political Searches in Social Media. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing, CSCW ’17*, 417–432. New York, NY, USA: ACM. ISBN 978-1-4503-4335-0. doi:10.1145/2998181.2998321. URL <http://doi.acm.org/10.1145/2998181.2998321>.

Kwak, H.; Lee, C.; Park, H.; and Moon, S. 2010. What is Twitter, a social network or a news media? In *Proceedings of the 19th International Conference on World Wide Web*, 591–600. ACM.

La Morgia, M.; Mei, A.; Mongardini, A. M.; and Stefa, J. 2023. Disinformation, Misinformation, and Polarization on Emerging Social Media: A Study on Bluesky. *arXiv preprint arXiv:2311.09984*.

Levy, R. 2021. Social media, news consumption, and polarization: Evidence from a field experiment. *American Economic Review* 111(3): 831–870.

Li, Y.; Fan, J.; Wang, Y.; and Tan, K.-L. 2018. Influence Maximization on Social Graphs: A Survey. In *IEEE Transactions on Knowledge and Data Engineering*, volume 30, 1852–1872. IEEE.

Lichterman, J. 2010. New Pew data: More Americans are getting news on Facebook and Twitter. <http://tinyurl.com/News-on-Social-Media>.

Liu, Z.; and Weber, I. 2014. Is Twitter a public sphere for online conflicts? A cross-ideological and cross-hierarchical look. In *International Conference on Social Informatics*, 336–347. Springer.

Lord, C. G.; Ross, L.; and Lepper, M. R. 1979. Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of personality and social psychology* 37(11): 2098.

Mellon, J.; Bailey, J.; Scott, R.; Breckwoldt, J.; and Miori, M. 2024. Do AI models produce human-like responses to political attitude questions? *arXiv preprint arXiv:2402.11005*.

Munson, S.; Lee, S.; and Resnick, P. 2013. Encouraging reading of diverse political viewpoints with a browser widget. In *Proceedings of the 7th International Conference on Weblogs and Social Media, ICWSM 2013*, 419–428. Boston, USA: AAAI press.

Newman, N.; Fletcher, R.; Robertson, C. T.; Eddy, K.; and Nielsen, R. K. 2023. Reuters Institute Digital News Report 2023. *Reuters Institute for the Study of Journalism* URL <https://reutersinstitute.politics.ox.ac.uk/digital-news-report/2023>.

Nyhan, B.; and Reifler, J. 2010. When corrections fail: The persistence of political misperceptions. *Political Behavior* 32(2): 303–330.

Pangakis, N.; Wolken, S.; and Fasching, N. 2023. Automated Annotation with Generative AI Requires Validation. *arXiv preprint arXiv:2306.00176*.

Pariser, E. 2011. *The Filter Bubble: What the Internet Is Hiding from You*. Penguin Group, The. ISBN 1594203008, 9781594203008.

Park, S.; Kang, S.; Chung, S.; and Song, J. 2009. NewsCube: delivering multiple aspects of news to mitigate media bias. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 443–452. ACM.

Pastor-Satorras, R.; and Vespignani, A. 2001. Epidemic spreading in scale-free networks. *Physical review letters* 86(14): 3200.

Pennycook, G.; and Rand, D. G. 2021. Shifting attention to accuracy can reduce misinformation online. *Nature* 592: 590–595.

Rauchfleisch, A.; and Kaiser, J. 2020. The False Positive Problem of Automatic Bot Detection in Social Science Research. In *PLOS ONE*, volume 15. Public Library of Science.

Ribeiro, F. N.; Saha, K.; Babaei, M.; Henrique, L.; Messias, J.; Benevenuto, F.; Goga, O.; Gummadi, K. P.; and Redmiles, E. M. 2019. On microtargeting socially divisive ads: A case study of russia-linked ad campaigns on facebook. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 140–149. ACM.

Richardson, M.; and Domingos, P. 2002. Mining knowledge-sharing sites for viral marketing. In *KDD*.

- Rizoiu, M.-A.; Graham, T.; Zhang, R.; Zhang, Y.; Ackland, R.; and Xie, L. 2018. # DebateNight: The Role and Influence of Socialbots on Twitter During the 1st 2016 US Presidential Debate. In *Twelfth International AAAI Conference on Web and Social Media*.
- Robertson, R. E.; Jiang, S.; Joseph, K.; Friedland, L.; Lazer, D.; and Wilson, C. 2018. Auditing Partisan Audience Bias within Google Search. *Proceedings of the ACM on Human-Computer Interaction* 2(CSCW).
- Santurkar, S.; Durmus, E.; Ladd, F.; Lee, E.; Liang, P.; and Hashimoto, T. 2023. Whose Opinions Do Language Models Reflect? .
- Schkade, D.; Sunstein, C. R.; and Hastie, R. 2007. What happened on deliberation day? *California Law Review* 95(3): 915–940.
- Settle, J. E. 2018. *Frenemies: How Social Media Polarizes America*. Cambridge University Press.
- Shoemaker, P. J.; and Vos, T. 2009. *Gatekeeping theory*. Routledge.
- Starbird, K.; Arif, A.; Wilson, T.; Van Koeveing, K.; Yefimova, K.; and Scarnecchia, D. 2018. Ecosystem or echo-system? Exploring content sharing across alternative media domains. In *Twelfth International AAAI Conference on Web and Social Media*.
- Sunstein, C. R. 2002. The Law of Group Polarization. *Journal of Political Philosophy* 10(2): 175–195. ISSN 1467-9760. doi:10.1111/1467-9760.00148. URL <http://dx.doi.org/10.1111/1467-9760.00148>.
- Taber, C. S.; and Lodge, M. 2006. Motivated skepticism in the evaluation of political beliefs. *American journal of political science* 50(3): 755–769.
- Teevan, J.; Ramage, D.; and Morris, M. R. 2011. # TwitterSearch: a comparison of microblog search and web search. In *WSDM '11 Proceedings of the fourth ACM international conference on Web search and data mining*, 35–44. Hong Kong, China: ACM.
- Törnberg, P. 2023. ChatGPT-4 outperforms experts and crowd workers in annotating political Twitter messages with zero-shot learning. *arXiv preprint arXiv:2304.06588* .
- Tsang, A.; Wilder, B.; Rice, E.; Tambe, M.; and Zick, Y. 2019a. Group-Fairness in Influence Maximization. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 5997–6005.
- Tsang, A.; Wilder, B.; Rice, E.; Tambe, M.; and Zick, Y. 2019b. Group-Fairness in Influence Maximization. *arXiv preprint arXiv:1903.00967* .
- Vosoughi, S.; Roy, D.; and Aral, S. 2018. The spread of true and false news online. *Science* 359(6380): 1146–1151.
- Vraga, E. K.; and Bode, L. 2017. Using expert sources to correct health misinformation in social media. *Science Communication* 39(5): 621–645.
- Wang, Z. T.; and Waseem, Z. 2021. Want to Reduce Labeling Cost? GPT-3 Can Help. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 4195–4205. Association for Computational Linguistics.
- Weber, P. 2012. The virtual get-together. Determinants interpersonal - "o public communication on news websites. *Social Media and Web Science. Frankfurt am Main: DGI* 457–459.
- Weber, P. 2014. Discussions in the comments section: Factors influencing participation and interactivity in online newspapers' reader comments. *New media & society* 16(6): 941–957.
- Wolsey, L. A. 1982. An analysis of the greedy algorithm for the submodular set covering problem. *Combinatorica* 2(4): 385–393.
- Wood, T.; and Porter, E. 2019. The elusive backfire effect: Mass attitudes' steadfast factual adherence. *Political Behavior* 41(1): 135–163.
- Xiao, Y.; Jiang, G.; Xu, W.; Saez-Trumper, D.; and Liao, Q. V. 2024. A Study of Political Discourse on Bluesky Social. *arXiv preprint arXiv:2405.18390* .
- Yarchi, M.; Baden, C.; and Kligler-Vilenchik, N. 2021. Political Polarization on the Digital Sphere: A Cross-platform, Over-time Analysis of Interactional, Positional, and Affective Polarization on Social Media. In *Political Communication*, volume 38, 98–139. Taylor & Francis.
- Ziems, C.; Held, W.; Shaikh, O.; Chen, J.; Zhang, Z.; and Yang, D. 2024. Can Large Language Models Transform Computational Social Science? *Computational Linguistics* 50(1): 237–291.